# Dr. Stefanos Georgiou

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**Location**: In Budapest but prefer remote positions

**Keywords**: Back-End, DevOps, Automations, Integrations,
and Energy & Run-Time Performance Researcher

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**About me**: I am passionated with research, **coding**, integrating new technologies, and automating cumbersome tasks. My research interests lie to Green and Energy-Efficient Computing, Mining Software Repositories, and Machine Learning. I enjoy working on the command-line and especially with Linux systems. I love to create small indepentant componets and tools to facilitate different daily functionalities. On my free time, I enjoy sporting, reading books, and **travelling**.

**NOTE**:Please consider wathcing the online CV that has diagrams and links https://stefanos1316.github.io/my_curriculum_vitae/index.html

**View as**: **[PDF] [DOCX] [RFT]**

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## Education

**2022-2024** **MSc, Palinka Maaster Consulting**; Hungarian University of Agriculture and Life Science *Still in progress

**2021-2022** **PostDoctoral, Electrical and Computer Engineering**; Queen's University Lab name: Software Evolution and Analytics Lab Lab's head: **Prof. Ying (Jenny) Zou**

**2016-2021** **PhD, Computer Science**; Athens University of Economics and Business Thesis title: *Energy and Run-Time Performance Practices in Software Engineering*; **Thesis** PhD advisors: (Main) **Prof. Diomidis Spinellis**, **Prof. Panos Louridas**, and **Rizos Sakellariou**

**2013-2015** **MSc, Pervasive Computing and Communications for Sustainable Development (PERCCOM)**; Erasmus Mundus Joint Master Degree ITMO University of Saint-Petersburg (Russia) Thesis title: *Implementating Green IT Approach for Transferring Big Data over Parallel Data Links*; **Thesis** Lulea University of Technology (Sweden) Semester 3: *Resource Efficient Pervasive Computing Systems and Communications* **Degree** Lappeenranta University of Technology (Finland) Semester 2: *Smart Software and Services* **Degree** Unversity of Lorraine (France) Semester 1: *Sustainable Computer Network Engineering* **Degree**

**2008-2013** **BSc, Networks and Systems Programming**; University of Cyprus **Degree** Thesis title: *Implementation and Evaluation of the Biologically – Inspired AntHocNet Routing Protocol in Sensor Network*; **Degree**

# Research Publications

- Conference Papers
  - Guoliang Zhao, **Stefanos Georgiou**, Safwat Hassan, Ying Zou, Derek Truong, Toby Corbin. Enhancing performance bug prediction using performance code metrics. In 21st International Conference on Mining Software Repositories: Technical Track, MSR '24.
  - **Stefanos Georgiou***, Maria Kechagia, Tushar Sharma, Federica Sarro, and Jenny Zou. Green AI: Do Deep Learning Frameworks Have Different Costs? In 44th International Conference on Software Engineering: Technical Track, ICSE '22. [**Paper**] [**BibTex**]
  - **Stefanos Georgiou**, Maria Kechagia Panos Louridas, and Diomidis Spinellis. What Are Your Programming Language's Energy-Delay Implications? In 15th International Conference on Mining Software Repositories: Technical Track, MSR '18. [**Paper**] [**BibTex**] [**Presentation**] [**Data-set**]
  - **Stefanos Georgiou**, Maria Kechagia, and Diomidis Spinellis. Analyzing Programming Languages' Energy Consumption: An Empirical Study. In PCI 2017: Proceedings of the 21st Pan-Hellenic Conference on Informatics, ACM International Conference Proceeding Series. ACM Press, September 2017. [**Paper**] [**BibTex**] [**Presentation**] [**Data-Set**]
- Journal Articles
  - Tushar Sharma, Maria Kechagia, **Stefanos Georgiou**, Rohit Tiwari, Indira Vats, Hadi Moazen, Federica Sarro. A survey on machine learning techniques for source code analysis. Accpeted for publication in Elsevier Journal of Systems and Software 2024.
  - Shayan Noei, Heng Li, **Stefanos Georgiou**, Ying Zou. An empirical study of refactoring rhythms and tactics in the software development process. Accepted for publication in Transactions on Software Engineering.
  - Maram Assi, Safwat Hassan, **Stefanos Georgiou**, Ying Zou. Predicting the change impact of resolving defects by leveraging the topics of issue reports in open source software systems. Accepted for publication in Transactions on Software Engineering and Methodology 2023.
  - Tushar Sharma, **Stefanos Georgiou**, Maria Kechagia, Taher A Ghaleb, Federica Sarro. Investigating developers' perception on software testability and its effects. Accepted for publication in Empirical Software Engineering 2023.
  - **Stefanos Georgiou**, Dimitris Mitropoulos, and Diomidis Spinellis. Energy-Efficient Computing in a Secure Environment. Submitted in July 2020 for review in ACM Transaction of Computer Systems.
  - **Stefanos Georgiou** and Diomidis Spinellis. Energy-Delay Investigation of Remote Inter-Process Communication Technologies. Accepted for publication in Elsevier Journal of Systems and Software 2019. [**Preprint**] [**BibTex**]
  - **Stefanos Georgiou**, Stamatia Rizou, and Diomidis Spinellis. Software Development Life Cycle for Energy-Efficiency: Techniques and Tools. Accepted for publication in ACM Computing Surveys 2019. [**Preprint**] [**BibTex**]

# Organization Services

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| ***Program Committee*** | EASE 2022 for the Research Track |
| | ISEC 2022 for the Doctoral Symposium |
| | ESEC/FSE 2021 for the Student Research Competition |

ASE 2020 for the International Workshop on Sustainable Software Engineering

***Reviewer*** Journals submitted to Software: Practices and Experience

Journals submitted to ACM Computing Surveys (CSUR)

Research papers submitted to ESEC/FSE

***Co-reviewer*** Conference papers submitted to ICSE-SEIP, MSR, OSS, ICPC, ICCS, SANER, FSE, ASE, ICMSE and SATToSE

# Invited Talks

**Tutorials** *How to improve your CI/CD process* presented on 5th of July 2019 at BALab of Athens University of Economics and Business in Athens, Greece. **Presentation**

*Travis CI with CV template tutorial* presented on 7th of June 2018 at ITMO Univesity of Saint Petersburg for PERCCOM Master students in Saint-Petersburg, Russia. **Presentation**

**Research** *Saving Energy in Software Development by Making the Right Choices* presented on the 4th of March 2022 at Sustainable Software Engineering lecture of TU Delft. **Presentation Slides**

*What are your programming language's energy-delay implications?* presented on the 12th of June 2018 at PERCCOM's Summer School in Lappeenranta, Finland. **Proof**

# Teaching Experience

Having weekly meetings with five PhD students and two Master students to assist and advise them on their research studies (at Queen's University from 2021 to 2023)

Assisting my PhD advisor at Programming II course (Winter Semester of 2016, 2017, and 2018) by giving tutorial and Lab sessions in Java [**Repository**]

Gave Java lectures and tutorials for the 1st and 2nd Coding Boot Camp in Athens, Greece (Oct. 2017 - May 2018) [**Repository**] [**Proof**]

# Developer Working Experience

**Jul 2022 – currently** **Back-end developer** @ **SimpleTechs** (Remote position)

- Developing applications with TypeScript and via **NestJS** frameworks.
- Using Postgres as database, GQL, and **Prisma client** to manage database access.
- Developed core services for **flexgold app**

**Feb 2021 – Jul 2022** **Back-end developer and DevOps:** @ **Borocard** (Remote position) Coordinating the development of a savings management mobile application.

- Developing server via **NodeJS** to host appliaction and manage its clients.
- Build project management on JIRA and Trello for external collaborators.
- Used **Stripe** and **Belvo** to manage user's savings.
- Developed crons to automate logs/configs storage on the cloud and database sanitization.
- Manage the AWS hosting and **Mongo Atlas** for production server.
- Developed Ansible scripts to manage the AWS resources remotely.
- Creating **pre-commit hooks** to maintaine source code quality.

**Feb 2021 – Feb 2022** **PostDoctoral, Electrical and Computer Engineering** Working as a **full time** researcher for the Software Analytics and Evolution Lab at Queen's University, Canada

- Conducted research studies on Green Software Development.
- Assisting PhD and Master students in their research projects.

**Dec 2020 – Feb 2021** **Software Developer** @ **Greek Free, Open Source Software** Developing a privacy-preserving epidemic dosimeter based on contact tracing (**full time**). **Proof Repository**

- Developing **Python**, **Shell**, and **Ansible** scripts on a Raspberry Pi Zero

**Jan 2019 – May 2020** **Back-end developer, DevOps, and Integrations:** @ **AllCanCode Inc. Greek Branch** Supported the product that facilitates fast web-sites development (by using **Blockly**) on Desktop that can be exported in smart-phones as well (**part time**).

- Development of API end-points in **Node.JS** and **MongoDB** for the product.
- Creation of the CI/CD pipeline for the product (in GitLab) to perform back-end and front-end testing (through **Mocha**, **Chai** and **Cypress** frameworks, respectively) and auto-deploy product on Google's App Engine (Aurora, Beta, and Production) using the **Flex environment**.
- Migration products functions to cloud functions using **Firebase hosting**, **Cloud Functions**, and **Cloud Run**.
- Integration of **Bitrise** system in the product to allow platform users to export their Desktop applications to Android and iOS through the **Cordova** wrapper.

**Jan 2016 – Jan 2019** **Proposals writing:** **Singular Logic S.A.** Proposals writing for the European Projects Department as **part time**. **Proof**

# TECH SKILLSET

***Back-End*** BASH and SHELL: often automating cumbersome and time consuming tasks

NodeJS: mainly using Express and prefer Mocha-Chai for testing

Java: was teaching it during my PhD

Python: No need to read tutorials, feeling confy to code anytime

C and C++: First languages to learn and impressed by their performance

MongoDB: Can use native but prefer Mongoose

Code Quality: Coveralls, NYC, ESLint, Prettier, and so on

Pre-commit hooks: Creating my own pre-commit hooks to ease boring tasks

***DevOps*** Ci/CD: Fan of Travis, GitLab, and GitHub Actions

Containers and Compose: to facilitate app deployment and execution

Hosting: used Firebase, AWS, and Google App Engine

Mobile CI/CD: Bitrise with Cordova wrapper for mobile apps

Configuration Management: knows Ansible because he enjoys YAML

***Certificates*** Unix Tools: **Data, Software, and Production Engineering**

MongoDB: **Basics**, **Aggregation**, **JavaScript**, and **Performance**

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Loading the data just for you.
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# RECOGNITIONS AND LANGUAGES

- Grands and awards:
  - **Arctic Code Vault Contributor**
  - Mentored younger researcher **who won 2nd place** in SRC ESEC/FSE '19.
  - Best paper award from the 15th Annual DMST Student Conference (2018)
  - Marie Skłodowska-Curie funds, **SENECA** (€ 62,400)
  - Erasmus Mundus Masters Scholarship, PERCCOM (€ 12,0000)
- Human Languages:
  - Greek/Hungarian (Native speaker)
  - English (Fluent speaker)
  - Russian/France (Basic knowledge)
- Living Abroad:
  - Born in Gyula, Hungary but grown up in Nicosia, Cyprus.
  - Ate baguettes and drunk wine in Lorraine, France (Fall Semester of 2013)
  - Attended to many sauna parties in Lappeenranta, Finland (Spring Semester of 2014)
  - Fish ice-cold beers from lakes in Lulea, Sweden (Fall Semeseter of 2014)
  - Drunk vodka and ate borsh in Saint-Petersburg, Russia (Spring Semester of 2015)
  - Teaching in Athens and visiting islands in Greece (Fall Semester of 2016 till now)
  - Biked in Delft, the Netherlands for my PhD secondment (Half month of Sept. 2017)

# My Academic Genealogy Tree

Zoom-in for more details